Data Modelling and Form Design for a Business Application in Zoho Creator

Patient Appointment Booking Form — Basic Fields Use Case

PRANAV SRINIVAS

241401068

1 Aim

To design and build a functional "Patient Appointment Booking Form" application in Zoho Creator that demonstrates the correct usage of all thirteen Basic Fields available in the Form Builder, while solving a real healthcare front-desk requirement of scheduling patient appointments in a structured manner.

The application allows a patient to submit personal identity, contact details, department preference, symptoms, existing conditions, preferred schedule, and consent for treatment through a single online form, and enables the front desk to view, filter, and confirm appointments through a dedicated report.

2 Procedure

Step 1: Create the application — A new Zoho Creator application named "Patient Appointment Booking" was created, and a form named Appointment_Booking_Form was added inside it.

Step 2: Design the form sections — The form was organised into four logical sections — Patient Details, Contact Details, Medical Details, and Scheduling & Consent — so the layout stays patient-friendly and easy to fill.

Step 3: Add all thirteen Basic Fields — Name, Email, Address, Phone, Single Line, Multi Line, Number, Date, Time, Drop Down, Radio, Multi Select, and Checkbox fields were dragged from the Basic Fields palette and mapped to Patient_Name, Email_Address, Residential_Address, Contact_Number, Referred_By, Symptoms_Description, Age, Preferred_Appointment_Date, Preferred_Appointment_Time, Department, Gender, Existing_Conditions, and Consent_For_Treatment respectively.

Step 4: Configure field-level validation — Mandatory checks were enabled on Patient Name, Phone, Age, Gender, Department, Preferred Appointment Date, and Consent; the Email field's format validation, the Phone field's India country-code format, the Number field's 0–120 range, and the Date field's "no past dates" rule were all switched on.

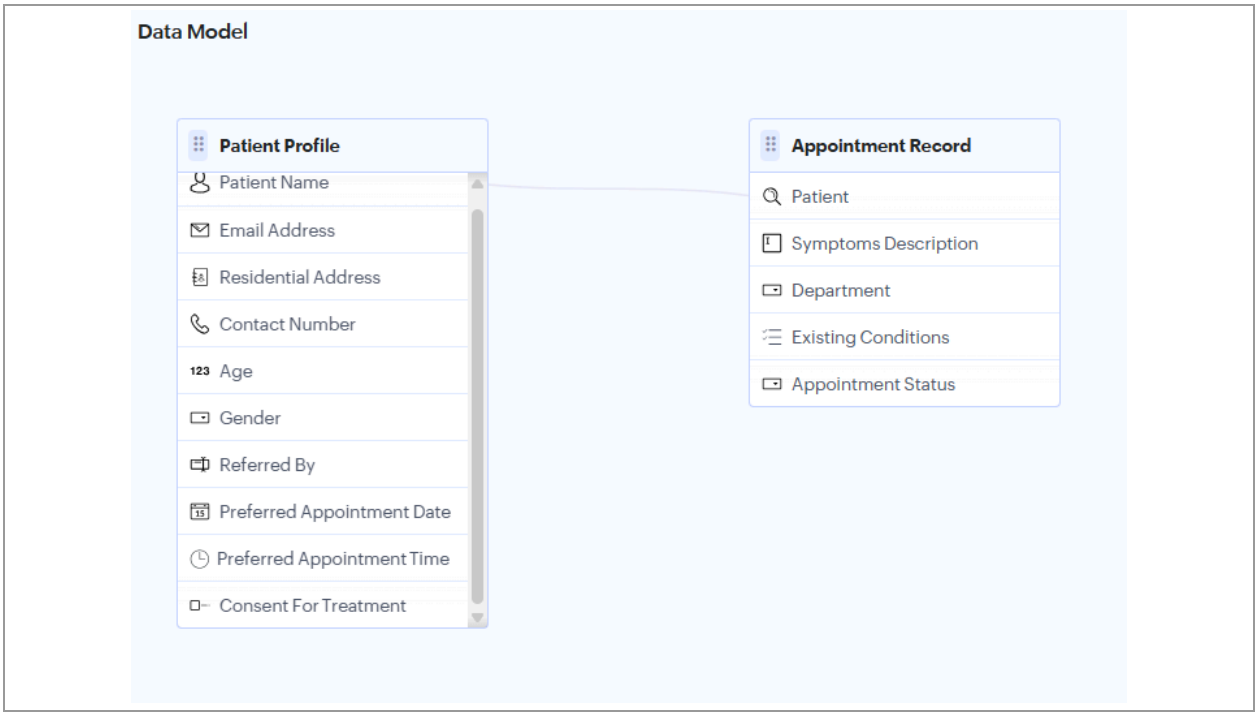
Step 5: Build the data model — A lookup relationship was created between the Patient Profile and the Appointment Record so that each appointment links back to the correct patient's stored details, keeping the data normalised.

Step 6: Automate workflows — Deluge-based workflows were configured to enforce the age limit on input, validate the preferred appointment date, send an automatic confirmation email on successful submission, and show a success alert once the record is saved.

Step 7: Preview and test the form — The live form was opened in Preview mode and filled with sample patient data to confirm that every field, section, and validation rule behaves as designed.

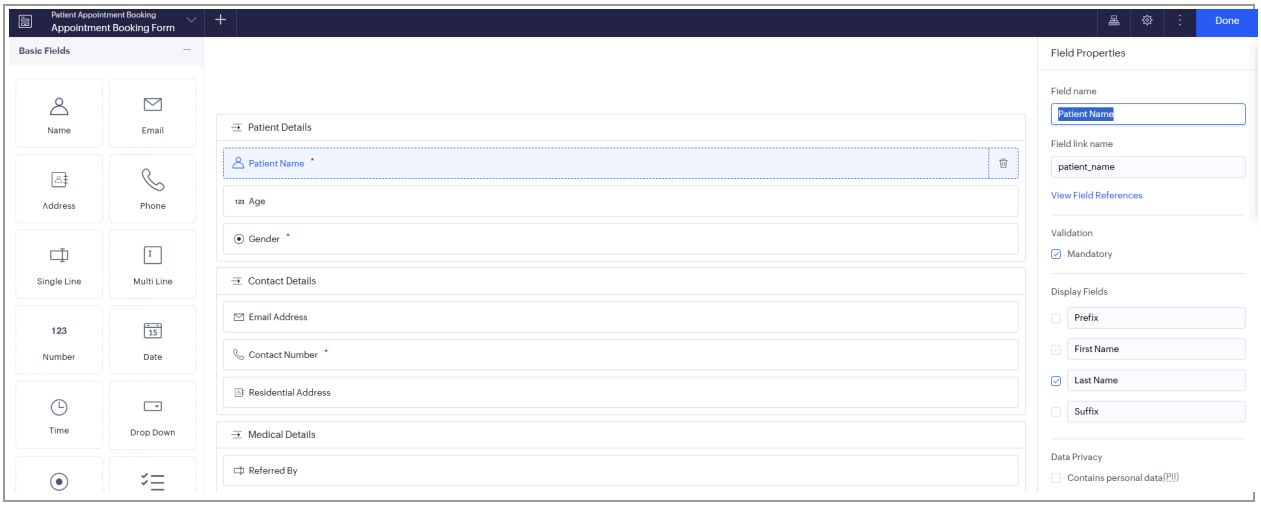
3 Output

31 Data Model — Patient Profile linked to Appointment Record



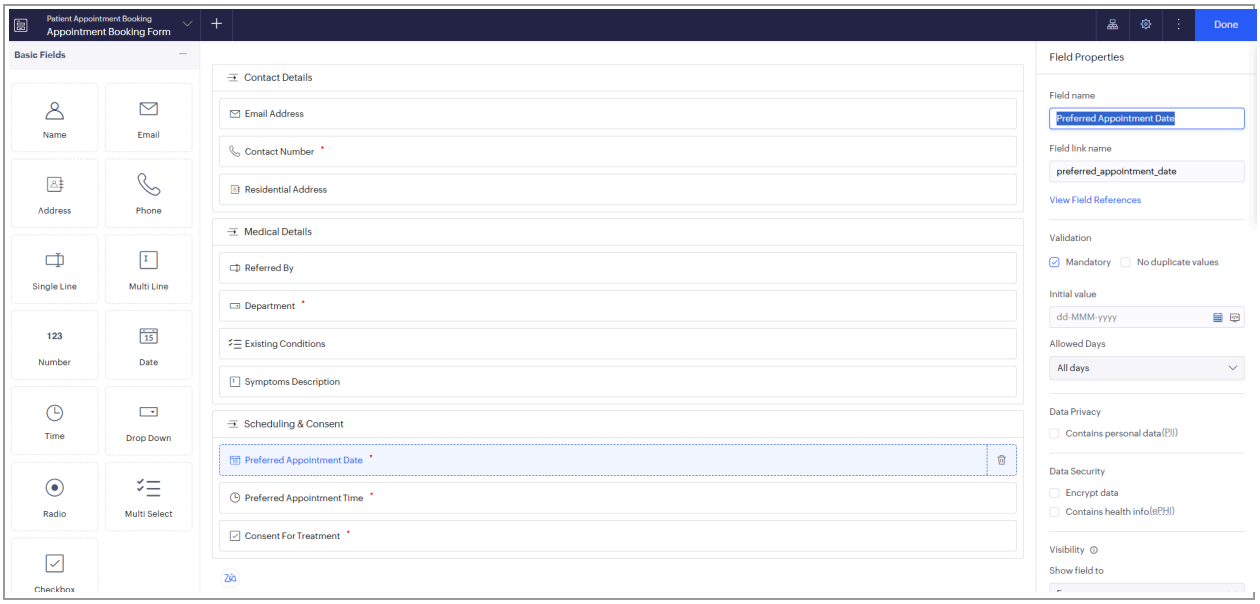
The data model shows the Patient Profile form holding all thirteen basic-field values, connected through a lookup to the Appointment Record, which stores the appointment-specific fields (Symptoms, Department, Existing Conditions, Status). This link lets one patient have multiple linked appointment records without duplicating personal data.

32 Form Builder — Basic Fields palette and Patient Details section



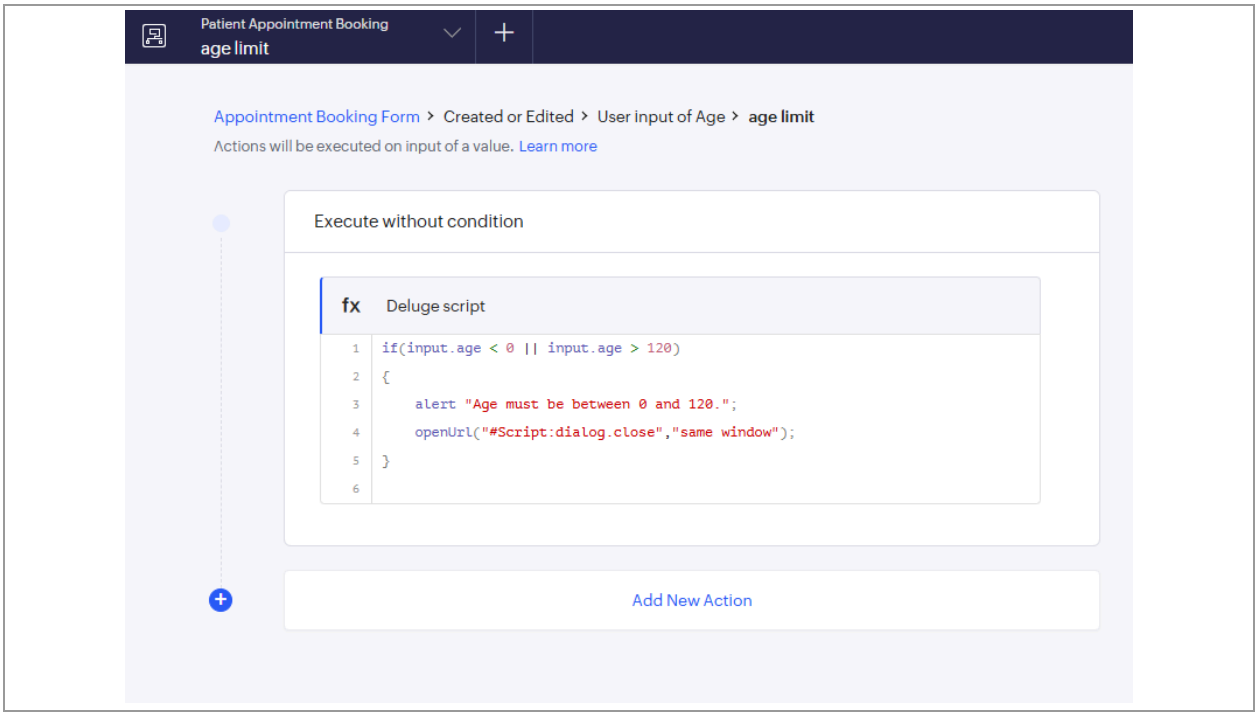
The Form Builder view shows the Basic Fields palette on the left and the Patient Name field selected in the Patient Details section, with its Field Properties panel confirming the Mandatory validation is switched on. The Contact Details and Medical Details sections are visible below, showing the section-wise layout in progress.

33 Form Builder — Preferred Appointment Date field configuration



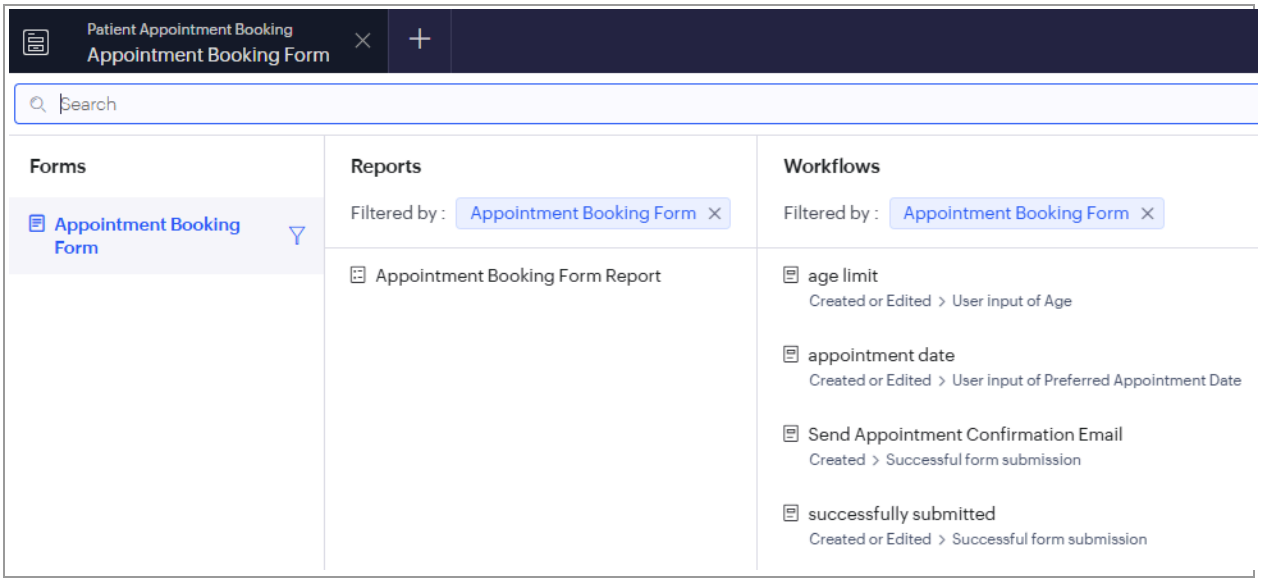
This view shows the Scheduling & Consent section with the Preferred Appointment Date field selected. The Field Properties panel on the right confirms the field is marked Mandatory and its Allowed Days is set, which is where the “no past dates” restriction is applied.

34 Workflow — Age limit validation script



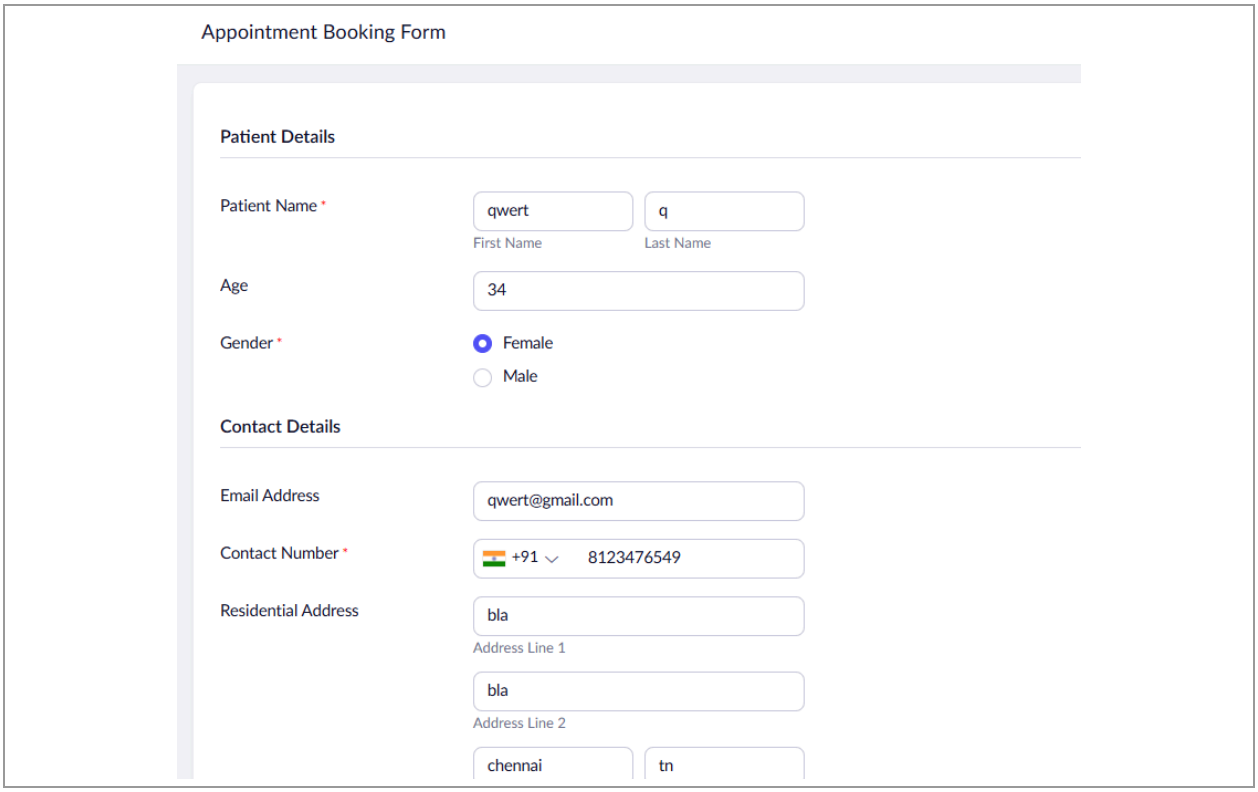
The “age limit” workflow, triggered on user input of Age, runs a Deluge script that checks whether the entered age falls outside 0–120. If it does, the script raises an alert (“Age must be between 0 and 120.”) and closes the dialog stopping an invalid value from being saved.

35 Application overview — Forms, Reports, and Workflows



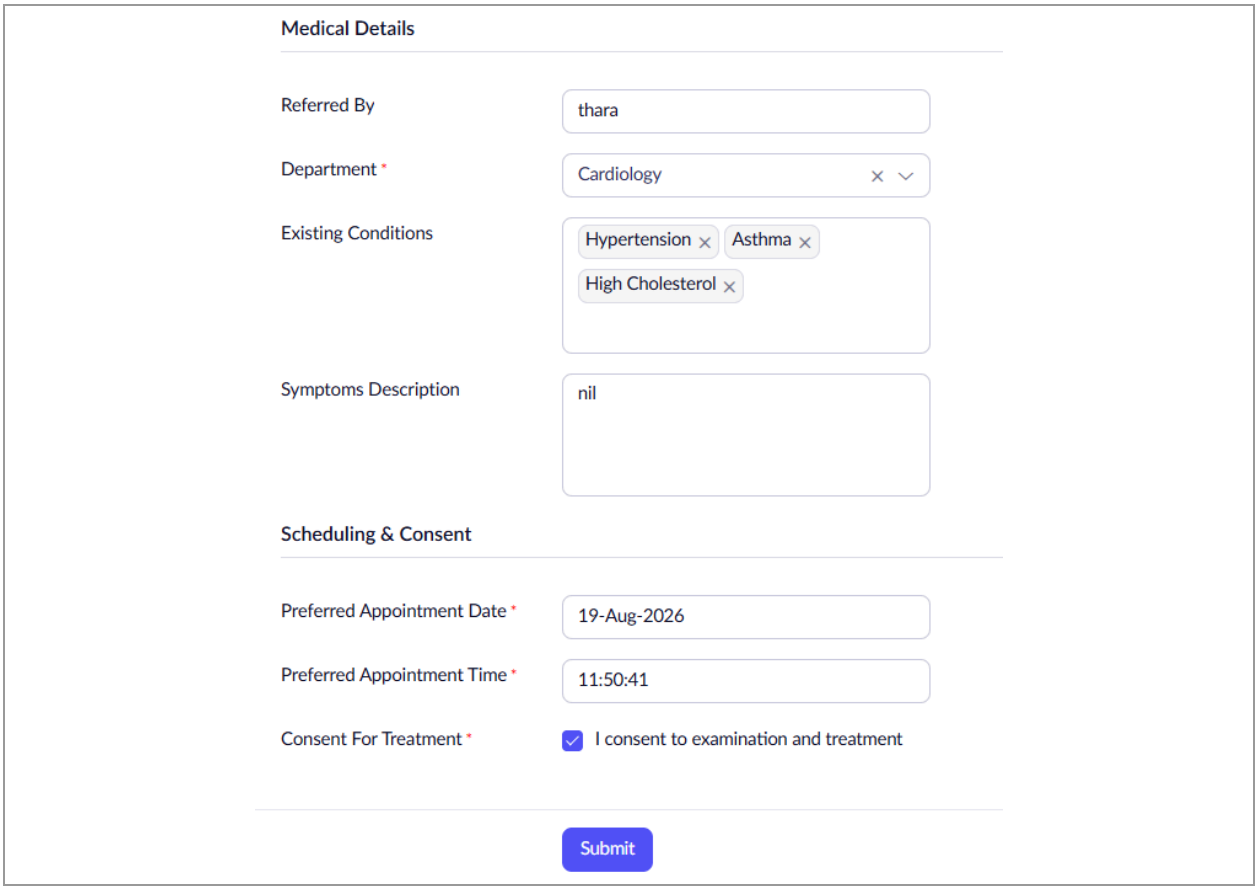
The application overview lists the Appointment_Booking_Form and its Appointment_Booking_Form Report, alongside four active workflows: age limit, appointment date, Send Appointment Confirmation Email, and successfully submitted. Together these automate validation and communication after every submission.

36 Live form preview — Patient Details and Contact Details



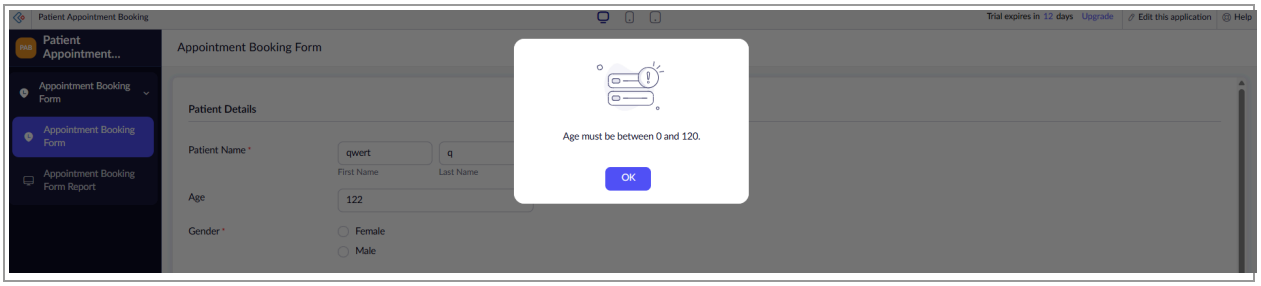
The live preview shows a sample submission with Patient Name, Age, Gender, Email Address, Contact Number (with India country code), and Residential Address (with city, state, pincode, and country) filled in, confirming these fields render and accept input correctly.

37 Live form preview — Medical Details, Scheduling and Consent



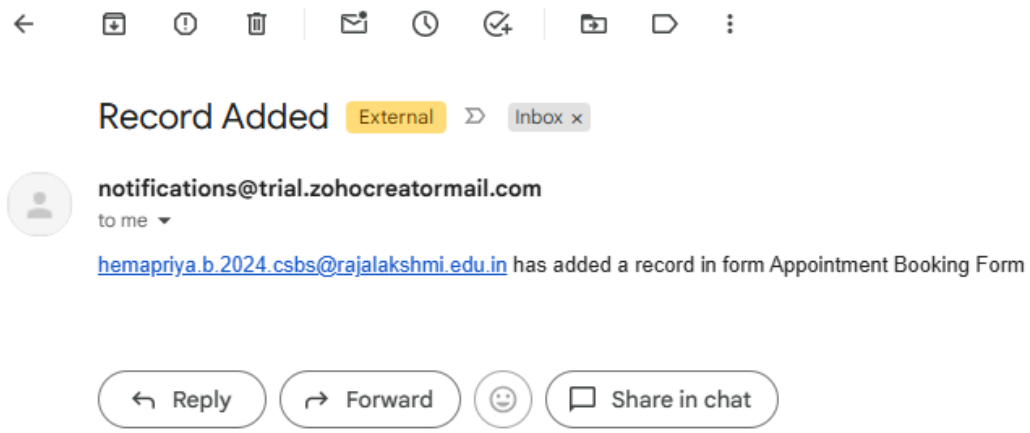
The second half of the same submission shows Referred By, Department, Existing Conditions (multi-select), and Symptoms Description, followed by the Preferred Appointment Date and Time and the Consent For Treatment checkbox, which must be checked before the Submit button can be used.

38 Validation in action — Age out of allowed range



Entering an age of 122 and attempting to submit triggers the Deluge validation script, which blocks submission and displays the alert “Age must be between 0 and 120.” This confirms the Number field's range validation and the age-limit workflow are both working as intended.

39 Appointment confirmation email (to be added)



4 Final Result

The Appointment_Booking_Form was successfully built in Zoho Creator using all thirteen Basic Fields, organised into four clear sections covering patient identity, contact information, medical details, and scheduling with consent.

Field-level validations — mandatory fields, email format, Indian phone format, the 0–120 age range, and the restriction on past appointment dates — were configured and verified to work correctly, including the tested error case for an out-of-range age.

A lookup-based data model links each Appointment Record back to its Patient Profile, and automated workflows enforce validation and trigger a confirmation email on successful submission, giving the front desk a structured, reliable way to review and confirm appointments through the Appointment_Booking_Report.

The exercise met its objective: every Basic Field type was correctly mapped to a real healthcare use case, and the resulting application replaces the clinic's manual, error-prone booking process with a validated, low-code digital form.